W-R-175/4D
22 December 1986
SUPERSEDING
W-R-175/4C
15 May 1973

FEDERAL SPECIFICATION SHEET

REELS, PRECISION, ALUMINUM, WITH 3-INCH CENTER HOLE

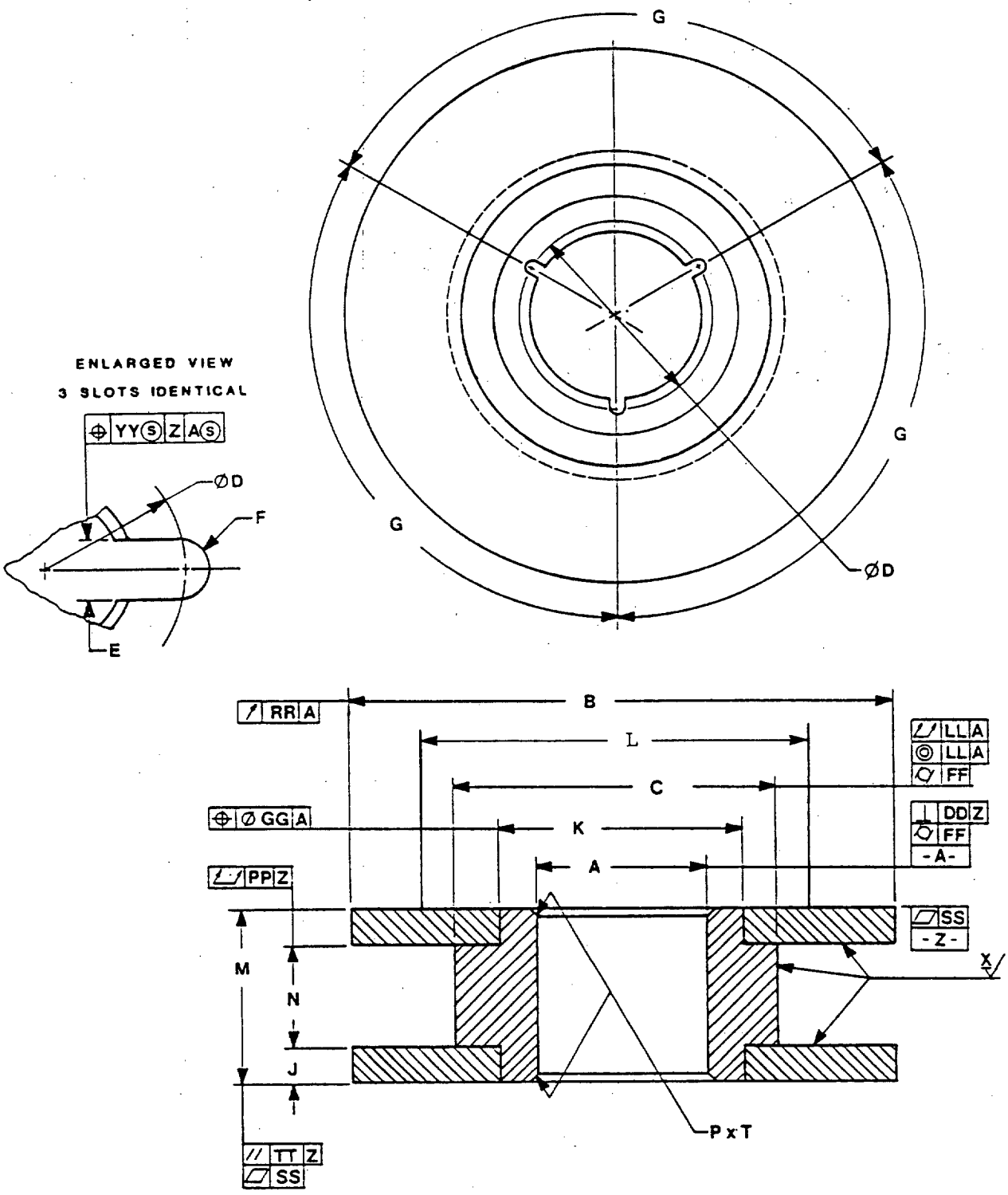
This specification sheet is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The requirements for acquiring the reels described herein shall consist of this specification and the latest issue of W-R-175.

AMSC N/A FSC 5835

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FIGURE 1. Reel configuration.

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REQUIREMENTS:

1. Materials: The reels shall be constructed of anodized aluminum or an alloy with aluminum as the principal element.

2. Flange fastening method: The reels shall have six fasteners per flange spaced 60 degrees plus or minus 12 minutes apart on a hole center diameter of 4.062 ± 0.005 inches. Flange fasteners shall be 6-32 by 5/16, carbon steel, electroplated, type III, style 2c machine screws as specified in FF-S-92. The thread depth of the hub flange mounting holes shall be not less than 0.25 inch. Flange fastening members shall be flush with or below outer surface of the flanges.

3. Reel dimensions: Unless otherwise specified, the reels shall be in accordance with the dimensions and measurements specified on figure 1 and in tables I and II.

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TABLE I. Dimensions.[1] [2]

Characteristic[3]	Reel and hub indicator	Dimensions		Flange deflection limit
		maximum	minimum	
Inside hub dia (A)		3.004 (76.302)	3.000 (76.200)	
Outside flange dia (B)	ECP	8.010 (203.45)	7.990 (202.95)	0.012 (0.305)
	JCP	9.760 (247.90)	9.740 (247.40)	.015 (0.381)
	PCP	10.510 (266.95)	10.490 (266.45)	.015 (0.381)
	QCP	12.510 (317.75)	12.490 (317.25)	.021 (0.533)
	SCP	14.010 (355.85)	13.990 (355.35)	.024 (0.610)
	TCP	15.010 (381.25)	14.990 (380.75)	.035 (0.889)
	UCP	16.010 (406.65)	15.990 (406.15)	.040 (1.016)
Outside hub dia (C)		4.510 (114.56)[4]	4.490 (114.05)	
Keyslot pitch dia (D)		3.252 (82.601)	3.248 (82.499)	
Keyslot width (E)		0.225 (5.715)	0.219 (5.563)	
Keyslot radius (F)		.5E	.5E	
Angle between successive keyslots (degrees) (G)		120.1 (2.096rad)	119.9 (2.093rad)	
Flange thickness (J)		0.098 (2.49)	0.084 (2.13)	
Inside diameter of flange dia[5] (K)		3.700 (93.980)[6]	3.600 (91.440)[6]	
Flange clearance between lateral mounting surfaces, dia[12] (L)			6.0 (152.4)	

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Tape width indicator	Dimensions		Flange deflection limit
		maximum	minimum	
Overall reel width within lateral mounting surface (M)	250	0.465 (11.811)	0.459 (11.659)	
	500	.715 (18.161)	.709 (18.009)	
	750	.965 (24.511)	.959 (24.359)	
	1000	1.215 (30.861)	1.209 (30.709)	
	1500	1.715 (43.561)	1.709 (43.409)	
	2000	2.215 (56.261)	2.209 (56.109)	
Distance between flanges at hub[7] (N)	250	0.272 (6.909)	0.268 (6.807)	
	500	.522 (13.259)	.518 (13.157)	
	750	.772 (19.609)	.768 (19.507)	
	1000	1.022 (25.959)	1.018 (25.857)	
	1500	1.522 (38.659)	1.518 (38.557)	
	2000	2.022 (51.359)	2.018 (51.257)	
Hub chamfer angle (degrees) (P)		46.0 (0.80rad)	44.0 (0.77rad)	
Hub chamfer (T)		0.040 (1.02)	0.020 (0.51)	
Mounting surface parallelism (TT)		.0014 (0.0356)		
Mounting surface flatness (SS)		.002 (0.051)		
Inside hub surface perpendicularity (DD)		.002 (0.051)[8]		
Inside hub surface cylindricity (FF)		.002 (0.051)[8]		

See footnotes at end of table.

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TABLE I. Dimensions.[1] [2] - Continued

Characteristic[3]	Dimensions		Flange deflection limit
	maximum	minimum	
Outside hub surface cylindricity (FF)	0.002 (0.051)[9]		
Outside hub surface concentricity (total runout) (LL)	.002 (0.051)		
Flange rim concentricity (RR)	.020 (0.51)		
Flange wobble (PP)	.005 (0.127)		
Keyslot position (YY)	.005 (0.127)		
Tape path clearance[10]			
Moment of inertia[11]			
Roughness of inside flange surface (X)	50 microinches		
Roughness of outside hub surface (X)	50 microinches		
Open flange area (percent)	25		
Diameter position (GG)	0.002 (0.051)		

See footnotes at end of table.

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[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] This dimension is exclusive of any friction ring used on the reel. If a friction ring is used it shall meet the following requirements:

(a) The material shall be neoprene of a durometer hardness of 60 ± 10 or it shall be a suitable elastomer.
(b) The outside diameter shall be concentric with the inside diameter within a tolerance of 0.020 (0.51).
(c) The inside and outside surfaces shall be cylindrical within a tolerance zone of 0.010 (0.25).
(d) The friction ring shall meet the temperature and humidity requirements of W-R-175.
(e) When mounted on the reel, the ring shall be centered between the flanges and its width shall be 0.020 to 0.040 (0.127 to 0.203) less than the applicable "N" dimension.
(f) The ring shall be flat on the hub and shall exhibit no visible buckling or puckering of the ring surface cracking or slitting when mounted on the reel.

[5] Lateral mounting surfaces extend out to "K" diameter.
[6] The "K" diameter for the flange shall be 3.700 maximum, and the "K" diameter for the hub shall be 3.600 minimum. The "K" diameter is included for reference purposes only. The "K" diameter will not be tested.
[7] The distance between the flanges, measured at the periphery of the flanges, shall not exceed the distance between the flanges at the hub (dimension "N") by more than the following applicable value for a given reel, measured within 0.130 inch from the flange periphery:

ECP - 0.020 (0.51)	SCP - 0.030 (0.76)
HCP - 0.020 (0.51)	TCP - 0.040 (1.02)
PCP - 0.026 (0.66)	UCP - 0.046 (1.17)
QCP - 0.030 (0.76)	

[8] The inside hub surface cylindricity and perpendicularity for reels for 0.250- and 0.500-inch tape shall be not greater than 0.0005 (0.0127).
[9] The outside hub surface cylindricity for reels for 0.250- and 0.500-inch tape shall be not greater than 0.0008 (0.0203).
[10] The tape path clearance shall be measured in accordance with W-R-175. The applicable tape path clearance shall be as follows:

Tape width indicator	Tape path center line
250	0.229 (5.817)
500	.355 (9.017)
750	.480 (12.192)
1000	.605 (15.367)
1500	.855 (21.717)
2000	1.105 (28.067)

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[11] See table II.
[12] "L" is the minimum diameter within which the flange, its fasteners and/or any labeling shall not protrude outside the planes of the lateral mounting surfaces.

TABLE II. Moments of inertia.[1]

Tape width indicator	Moment of inertia (max)				
	PCP	QCP	SCP	TCP	UCP
250	0.16(6.7)	0.33(13.9)	0.49(20.6)	0.62(26.1)	0.79(33.3)
500	.16(6.7)	.33(13.9)	.49(20.6)	.63(26.5)	.80(33.7)
750	.17(7.2)	.34(14.3)	.50(21.1)	.65(27.4)	.82(34.6)
1000	.17(7.2)	.34(14.3)	.51(21.5)	.66(27.8)	.83(35.0)
1500	.18(7.7)	.35(15.0)	.59(24.8)	.69(29.6)	.85(36.4)
2000	.22(9.3)	.39(16.4)	.62(26.1)	.71(29.9)	.87(36.7)

[1] Units are pound-feet squared. Metric equivalents (gram-meter squared) are included for information only.

QUALIFICATION:

1. The qualification sample for type I reels shall consist of ER-1000-SCP reels with friction rings. The qualification sample for type II reels shall consist of ER-1000-SCP reels with friction rings. Qualification of a particular reel type and configuration shall constitute qualification for all reels of that particular type and configuration regardless of reel width and flange diameter.

MILITARY INTERESTS:

Custodians

Army - CR
Navy - SH
Air Force - 99

Review activities

Navy - NV, OM
Air Force - 80
DLA - ES

User activity

Navy - AS

CIVIL COORDINATING ACTIVITY:
NASA - NPPO

Preparing activity:
Navy - SH
(Project 5835-0079-4)

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